DESIGN OF WIDEBAND CIRCULARLY POLARISED MICROSTRIP ANTENNA WITH A DEFECTED GROUND STRUCTURE

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Abstract -- **The fast-paced growth in wireless communication technologies, especially the coming of age of 5G infrastructure in mid-band spectrum (3.5 GHz), has created the need for very small-chunky, low-cost, and high-performance antennas that can support wideband operation with stable polarisation properties. CP antennas are desired as they provide relief from multipath fading, orientation mismatch, or atmospheric depolarisation. However, traditional microstrip patch antennas do not have good impedance bandwidth and axial ratio bandwidth (ARBW), so under wideband conditions, big limitations are set on their performance. The truncation of the patch corners excites two orthogonal modes with a 90° phase difference to generate circular polarisation. The dumbbell-shaped DGS etched into the ground plane improves the bandwidth and axial ratio characteristics by modifying the surface current distribution and suppressing unwanted modes. The antenna is fabricated on an FR4 substrate and simulated using CST M.**

Keywords– Circular Polarisation, Microstrip Patch Antenna, Defected Ground Structure (DGS), 5G Mid-Band, CST Microwave Studio

I. INTRODUCTION

Antennas are the essential part of wireless communication systems of today, and the way they are made or designed is a pivotal factor in their ability to operate in an efficient, reliable, and adaptable way. The launch of 5G technology has created an urgent demand for the antenna that will not only be in small size and cheap in price but will still be able to provide a very good performance in terms of the bandwidth, gain, and polarisation purity.

One of the major advancements in technology that have settled the CP antennas in the good place of the wireless communication systems is the fact that they are very reliable and not susceptible to the losses induced by the factors like orientation mismatch and multipath fading, which affect other types of antennas. Besides that, their polarisation is not much affected by the surrounding objects. The angular displacement of mobile devices is also not a prominent cause of errors for such CP antennas, and, therefore, they are well suited for the satellite communications realm, the RFID market, IoT, radar, and 5G wireless networks.More specifically, microstrip patch antennas are very popular for those same use cases because of their low profile, light weight, ease of integration with printed circuits, and low fabrication cost. Yet, their relatively low impedance bandwidth (usually 2–5%) and very limited ARBW are the main drawbacks. Their narrowband nature is due to the resonant cavity structure of a patch which is not easily adapted to wideband systems, such as 5G, where the requirements are strong.Some of the investigated methods to overcome these problems include Stacked patches and multilayer substrates, which allow for bandwidth increase but at the expense of added complexity and manufacturing cost.

Metamaterial inclusions, one of the promising ways to achieve performance enhancement but could be weakened by design complexity and loss mechanisms.

Defected Ground Structures (DGS) which, by changing the distribution of the current on the ground plane, offer a simple and low-cost solution for the increase of the bandwidth and axial ratio performance.From the list, DGS has turned out to be the most favourable option for improving the antenna performance to a great extent without expanding the size or making it more expensive. Various DGS concepts such as slots, rings, and dumbbells have been examined, each of which is able to make a specific contribution to impedance and polarisation improvements.

This research is centered around the dumbbell-shape defected ground structure integration with a truncated square circularly polarised microstrip patch antenna. The circular polarisation is, actually, a result of the truncation method, whereas the DGS serves to improve the impedance matching and to increase the ARBW. This is, indeed, a very good use of these two methods for the creation of a compact, efficient, and wideband antenna that is optimised for 3.5 GHz mid-band 5G applications.

II. LITERATURE REVIEW

This literature review synthesizes prior work on circularly polarised (CP) microstrip antennas, focusing on techniques that improve axial-ratio bandwidth (ARBW) and impedance bandwidth—particularly defected ground structures (DGS), truncation/perturbation single-feed methods, and geometry/slot-based optimizations. The discussion centers on the four papers you provided and situates them within the broader technical approaches used to achieve wideband CP performance.

The design of circularly polarized (CP) microstrip antennas has been widely investigated due to their robustness against multipath fading, orientation mismatch, and depolarization effects, making them attractive for applications ranging from satellite communications and RFID to emerging 5G networks. Across prior works, the main performance metrics remain consistent: return loss (S11), impedance bandwidth (IBW), axial ratio (≤ 3 dB for CP), axial-ratio bandwidth (ARBW), gain, directivity, radiation patterns, and efficiency. For 5G mid-band systems (3.3–3.8 GHz), antennas must simultaneously achieve wide ARBW and IBW while maintaining compactness, low cost, and ease of fabrication.

Early research emphasized *truncation and perturbation methods* for generating CP with a single feed. In square or rectangular patches, truncated corners excite two orthogonal modes with equal amplitude and 90° phase shift, a simple and manufacturable approach [5]. Perturbation slots and offset feeding have also been shown to achieve CP without hybrid couplers, though their ARBW remains narrow, typically only a few percent.

To address bandwidth limitations, researchers explored defected ground structures (DGS). By etching patterns such as dumbbell, ring, or plus-shaped slots into the ground plane, the surface current distribution is altered, effectively improving IBW and in some cases ARBW. However, DGS introduces design tradeoffs, including increased back radiation and fabrication complexity [6].

More recent works highlight the combination of these techniques. Lahmissi and Tounsi [1] developed a CP patch array for 5G, achieving high gain and beam steering through optimized element spacing. While directivity improved, ARBW remained narrow and mutual coupling effects limited robustness. Saha et al. [2] applied a plus-shaped DGS to a CP MIMO antenna at 2.4 GHz, enhancing impedance bandwidth and isolation. Despite success in ARBW and isolation, the design was confined to ISM bands and suffered from fabrication challenges, with limited scalability to 3.5 GHz. Sridhar and Poonkuzhali [3] proposed a CPW-fed dual-band CP antenna for WLAN and X-band.

Almeida et al. [4] explored ultra-wideband (UWB) CP using advanced patch shaping and multiple slot topologies, producing wide ARBW by merging resonances. While effective, UWB designs face efficiency issues on lossy substrates such as FR4 and are difficult to confine around 3.5 GHz without complexity. Complementing these, geometry-based slot and perturbation methods [7] offer wideband operation with parametric optimization but often require intricate designs that trade manufacturability for bandwidth.

Performance goals and common metrics

The latest investigations into circularly polarized (CP) microstrip antennas analyze performance based on the same metrics: return loss (S11), impedance bandwidth (usually defined at S11 < 10 dB), axial ratio (≤ 3 dB for acceptable CP), axial-ratio bandwidth (ARBW), realized gain, radiation patterns (co-polarized and cross-polarized), and overall efficiency. For applicable mid-band 5G applications (3.3–3.8 GHz), designs will need to achieve sufficiently wide ARBW to accommodate drift in frequency, tolerances in manufacture, and detuning due to environmental effects while keeping the design compact, with fabrication simplicity, and the most preferable design being single feed. In this work, CP performance is analyzed based on the results from the seventh design iteration of a simple microstrip patch antenna, where the length of the substrate was 28.8 mm, the length of the patch was 19.2 mm, and the patch feed width was 2.2 mm.

III. METHODOLOGY

1. Simple Mirco-strip Patch antenna for the target frequency

A rectangular/square microstrip patch radiates because its edges support standing waves (resonant cavity). Circular polarization (CP) with a single feed is commonly obtained by perturbing the square/rectangular patch so that two orthogonal modes of nearly equal amplitude are excited with a 90° phase difference. Corner truncation is a simple, compact perturbation that accomplishes this: cutting small symmetric corners shifts the two orthogonal resonances so that they are close in frequency and their phase difference at the operating frequency becomes 90°, producing RHCP or LHCP depending on the perturbation sense. The aim here is a compact, single-feed patch centered at 3.5 GHz with good impedance matching and stable CP performance.

Key Parameters:
Operating Frequency (fc): 3.5 GHz
Frequency Band: S-band (1–2 GHz)
Substrate Material: FR4(lossy)
$\varepsilon r = 4.3$, loss tangent ≈ 0.025
Substrate Thickness (h): 1.6 mm

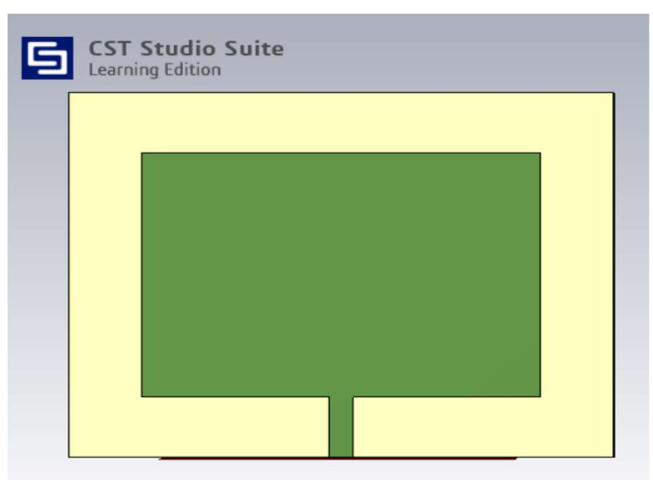


Figure 1. Design in CST Studio

Design Equations

(i) Patch Width (W)

The width of the patch is a key parameter that influences the impedance and radiation pattern. A good starting point is calculated using the following formula:

$$W = \frac{c}{2f_r}\sqrt{\frac{2}{\epsilon_r + 1}}$$

(ii) Effective Dielectric Constant (ϵ reff)

The patch radiates in a medium that is part substrate and part air. The effective dielectric constant accounts for this "fringing effect" and is always less than ϵ r.

$$\epsilon_{reff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2}\left[1 + 12\frac{h}{W}\right]^{-1/2}$$

(iii) Length Extension (ΔL)

First, calculate the extension in length due to the fringing fields at the two radiating edges.

$$\Delta L = 0.412h\frac{(\epsilon_{reff} + 0.3)\left(\frac{W}{h} + 0.264\right)}{(\epsilon_{reff} - 0.258)\left(\frac{W}{h} + 0.8\right)}$$

(iv) Patch Length (L)

The actual physical length of the patch is slightly less than half a wavelength in the effective dielectric medium.

$$L = \frac{c}{2f_r\sqrt{\epsilon_{reff}}} - 2\Delta L$$

(v) Calculation of Substrate Dimensions (Ws, Ls)

$$L_g = L + 6h$$

$$W_g = W + 6h$$

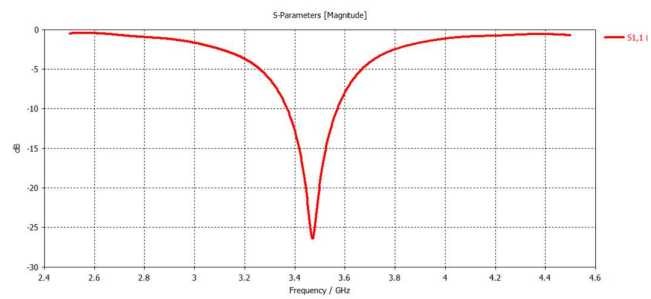
where h is the substrate heigh

t

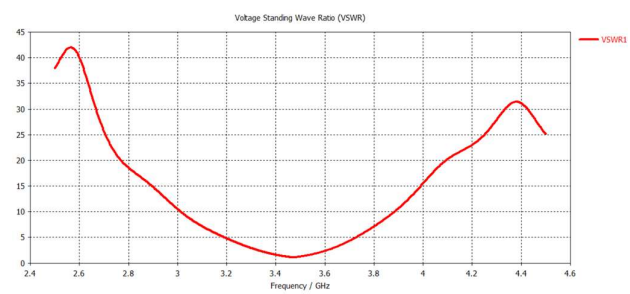
Specifications	Value
Substrate **Width (Ws)**	35.9 mm
Substrate **Length (Ls)**	29 mm
Patch **Length (Lp)**	19.4 mm
Patch **Width (Wp)**	26.3 mm
Feed line **Width (Wf)**	3.08 mm
Thickness of the Patch, Ground (**T**)	0.035 mm
Substrate Height (**h**)	1.6 mm

Table 1. Design dimensions for Microstrip Patch Antenna

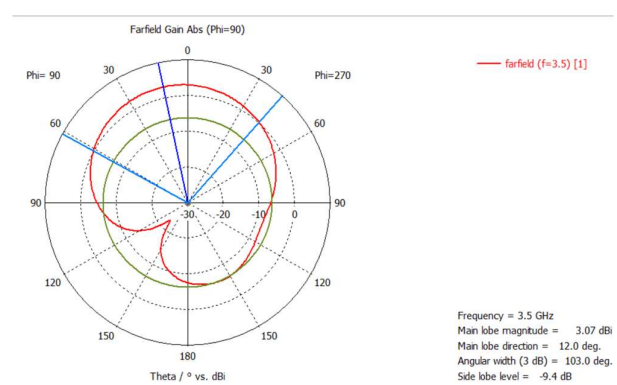
Result Observations (Microstrip Patch Antenna):



Graph 1. Return Loss (S11) vs Frequency



Graph 2. VSWR vs Frequency



Graph 3. Gain 2D Plot

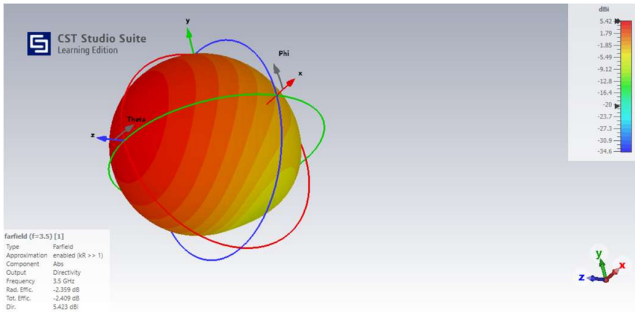


Figure 2. Gain 3D Plot

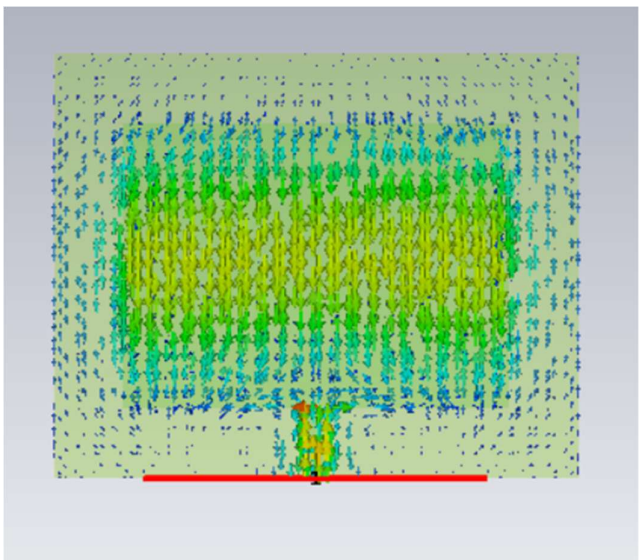


Figure 3. Surface Current Distribution

2. Truncated-square patch and Circularly Polarised antenna (with full ground)

Single-feed circular polarization (CP) typically occurs by simultaneously exciting two orthogonal modes with equal amplitudes and a 90° relative phase. One straightforward and compact solution is achieved by creating truncated corners on square or rectangular patches; in this scheme, the truncation length regulates the mode splitting, and centers the axial ratio (AR) frequency. Another method of manipulating mode amplitudes and phase involves placing perturbation slots or notches along patch edges. An alternative solution for exciting orthogonal modes is through offset feeding, where the feed point is moved away from the center of the patch. These techniques are valuable for their avoidance of hybrid couplers or dual-feed configurations, while being compatible with standard PCB fabrication.

Circular Polarisation using Truncated Corner Method (varying truncation length)

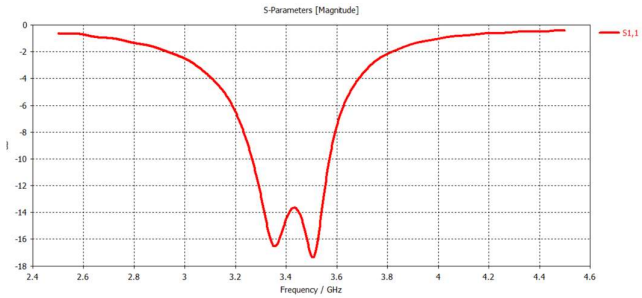
$$t \approx (0.03 \text{ to } 0.18) \times L$$

Specifications	Value
Substrate **Width (Ws)**	29.6 mm
Substrate **Length (Ls)**	30.5 mm
Patch **Length (Lp)**	20.9 mm
Patch **Width (Wp)**	20 mm
Feed line **Width (Wf)**	1.6 mm
Thickness of the Patch, Ground (**T**)	0.035 mm
Substrate Height (**h**)	3.6 mm
Truncation Length (**t**)	0.17*20.9= 3.55mm

Table 2. Optimal dimensions for the CP Antenna

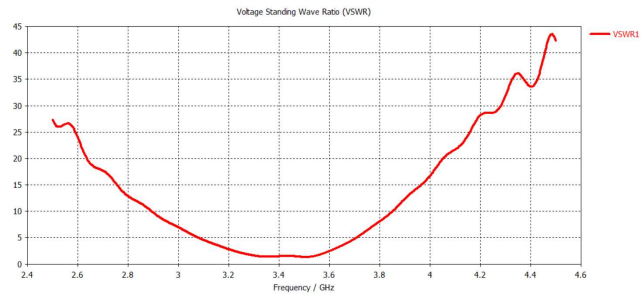
Figure 4. Design of CP antenna in CST studio

Result Observations (Circular Polarisation):



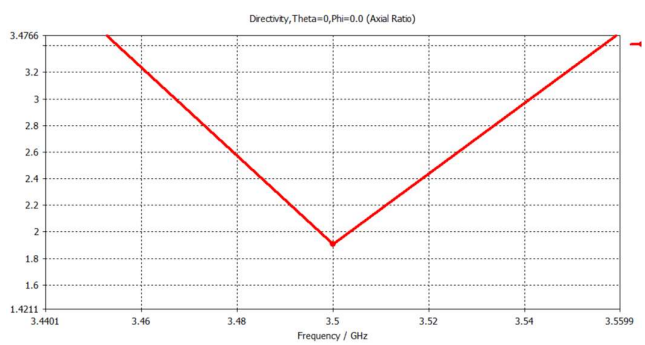
Graph 4. Return Loss(S11) vs Frequency

From the graph 4, it is observed that the return loss is less than 10dB for a Bandwidth of 304 MHz, and at 3.5 GHz it measures -17.25 dB.



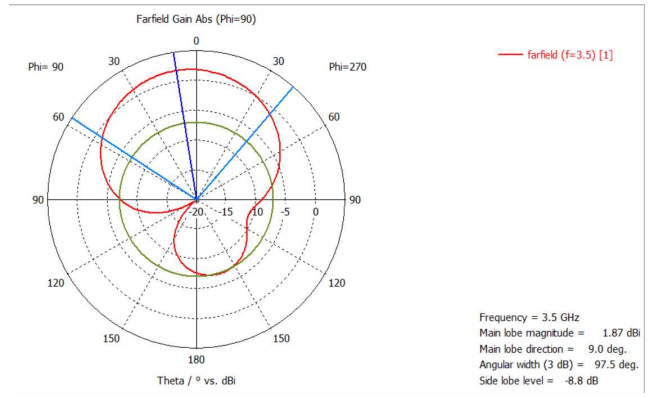
Graph 5. VSWR vs Frequency

From the graph 5, it is observed that VSWR is optimal (<2) for the same bandwidth.



Graph 6. Axial Ratio vs Frequency

The Axial Ratio remains below 3 dB (AR < 3dB) for a bandwidth of 74 MHz meaning that the circular polarization in the antenna is good.



Graph 7. Gain 2D Plot

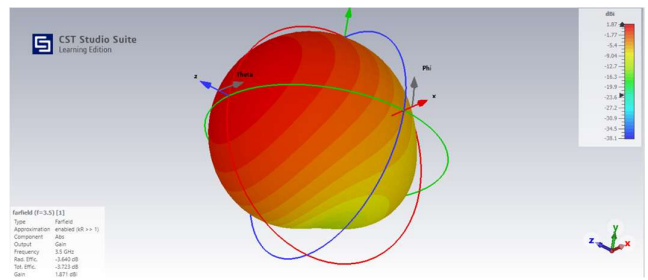


Figure 5. Gain 3D Plot

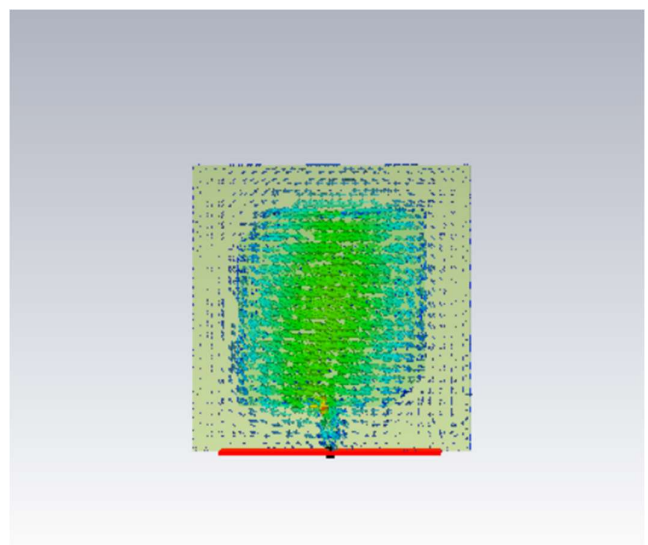


Figure 6. Surface Current Distribution

3. Introduction of Defected Ground Structure (DGS): role and design tradeoffs

In a defected ground structure (DGS), the distribution of current over the ground-plane is altered and changes the effective inductance and capacitance that the patch experiences. These changes can lead to better bandwidth, in terms of both impedance bandwidth, and in certain cases, axial-ratio bandwidth due to the resonant behavior generated from the ground slot. DGS is also helpful in suppressing spurious modes and improving impedance matching, as it has bandstop or low-pass filtering characteristics. Additionally, the loading effect of DGS can allow for antenna miniaturization due to a decreased physical size. However, higher bandwidths or filtering characteristics come with tradeoffs. In particular, DGS may potentially increase back radiation, decrease symmetry in radiation patterns, and increase sensitivity to substrate coupling. Lastly, complex slot geometries can add challenges to fabrication, making parametric studies to optimize these parameters related to slot length, width, and spacing an important element of the design and simulation effort.

Design: For this antenna, dumbbell shaped DGS has been used with the specifications for optimised results.

L_{dumb}=7.5mm

W_{dumb}=1.3mm

R_{disc}=1.2mm

Defective Ground Structure(DGS):

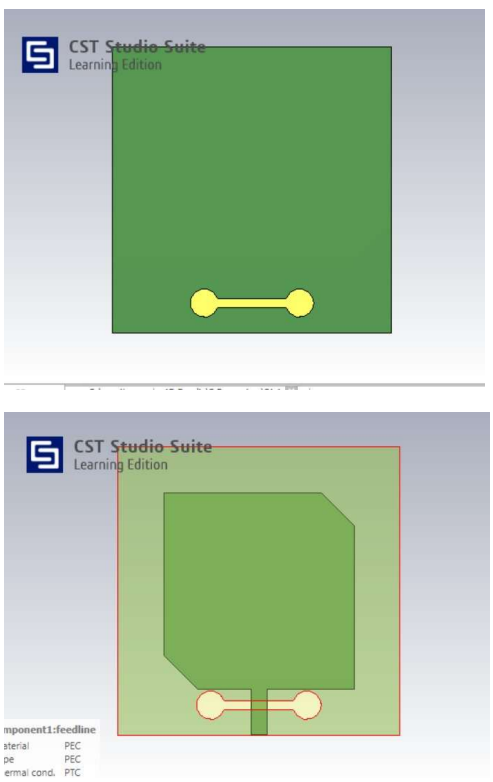
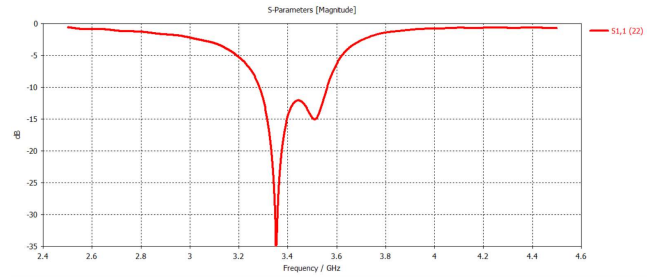
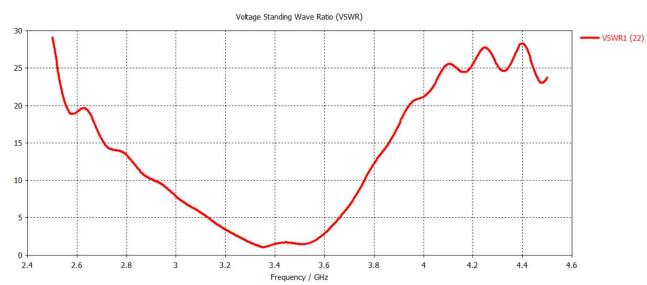


Figure 8. Desing in CST



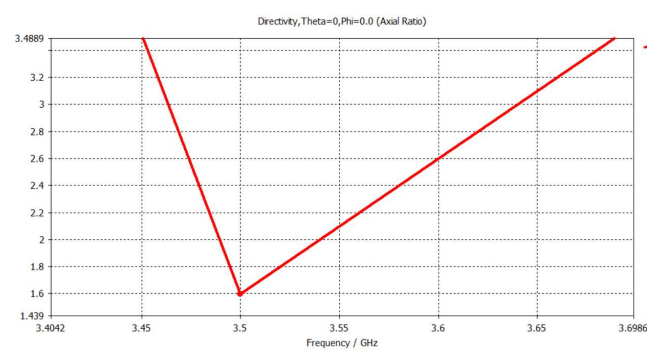
Graph 8. Return Loss vs Frequency

From the graph 4, it is observed that the return loss is less than 10dB for a Bandwidth of 278 MHz, and at 3.5 GHz it measures -14.78 dB.



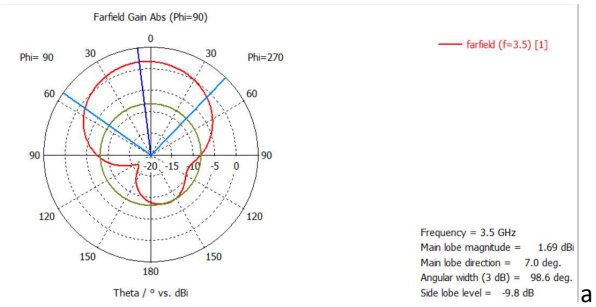
Graph 9. VSWR vs Frequency

From the graph 5, it is observed that VSWR is optimal (<2) for the same bandwidth.



Graph 10. Axial Ratio vs Frequency

The Axial Ratio remains below 3 dB (AR < 3dB) for a bandwidth of 101 MHz meaning that the circular polarization in the antenna is good. The ARBW has improved after the introduction of the defected ground structure.



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Graph 11. Gain 2D Plot

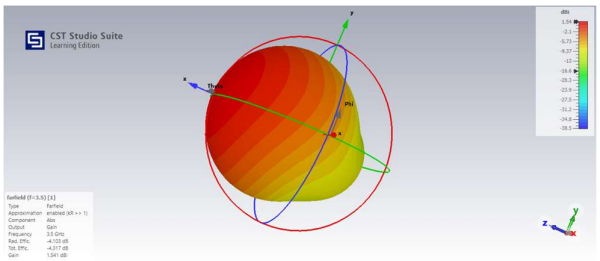

Figure 9. .Gain 3D Plot

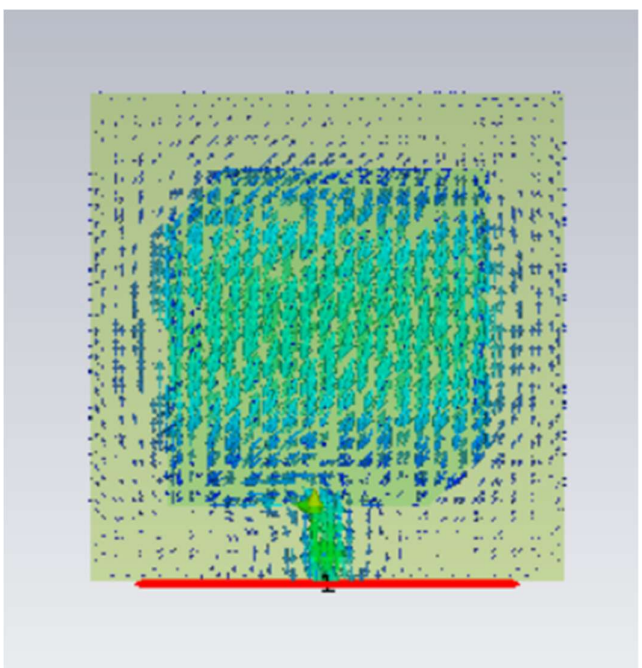

Figure 10. Surface Current Distribution

IV. RESULTS AND COMPARISON

All dimensions are kept the same for comparison. The dimensions of the dumbbell shape were swept and the best ones were considered for comparison.

Table 3. Comparison of Antenna parameters for a circularly polarised antenna without and with a defected ground structure (DGS)- from this work

Parameter	**Without DGS**	**With DGS**
Return Loss (S11) at 3.5 GHz	-17.25 dB	-14.78 dB
Impedance Bandwidth	304 MHz	278 MHz
Axial Ratio (AR) At 3.5 GHz	1.90 dB	1.59 dB
Axial Ratio Bandwidth (ARBW)	74 MHz	101 MHz
Directivity (dB)	5.51 Db	5.62 dB
Gain (dB)	1.87 Db	1.69 dB
Realised Gain (dB)	1.79dB	1.54dB

The introduction of the DGS led to a significant improvement in the antenna's circular polarization, primarily by increasing the axial ratio bandwidth by 36.5% and lowering the axial ratio value at 3.5 GHz. This enhancement, however, resulted in a trade-off, slightly degrading the impedance bandwidth and reducing the realized gain.

V. REFERENCES

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